

CIS 613: Technical Reviews

1

Outline

- Economics of Software Reviews
- Roles and Responsibilities in a Review
 - Responsible designer
 - Recorder
 - Review leader
 - Reviewer
- Types of Reviews
- An Industrial-Strength Review Process
- Effective Review Culture

2

About this Chapter / Topic

- Based on 20 years experience with an extremely mature software review process (Jorgensen at GTE)
 - Used in the development of telephone switching system software
 - High reliability and performance characteristics
- Many details developed “from experience”
- Matches my own experience (12+ years in Aerospace)

A question for the end: should reviews in high-assurance fields match reviews in other fields?

3

What are Technical Reviews

- A critical evaluation of a product
- A scheduled activity with a budget, inputs, and outputs
- A business meeting
- A group (or social) process
- Things that occur in everyday life

4

Reviews in Everyday Life

- Many forms of reviews are common
 - annual **physical** examination
 - **building code** inspection of a new home
 - film **reviews**
 - restaurant **reviews**
- In each of these, we can identify tasks/roles that review participants perform
- For each role, we can observe varying amounts of preparation

What is it about a review that helps us accept its conclusions?

5

What do Successful Reviews Need

- Process credibility
- Reviewer competence
- Reviewer preparation
- Mechanics / process

6

In Class Assignment: Reasons to Have Reviews

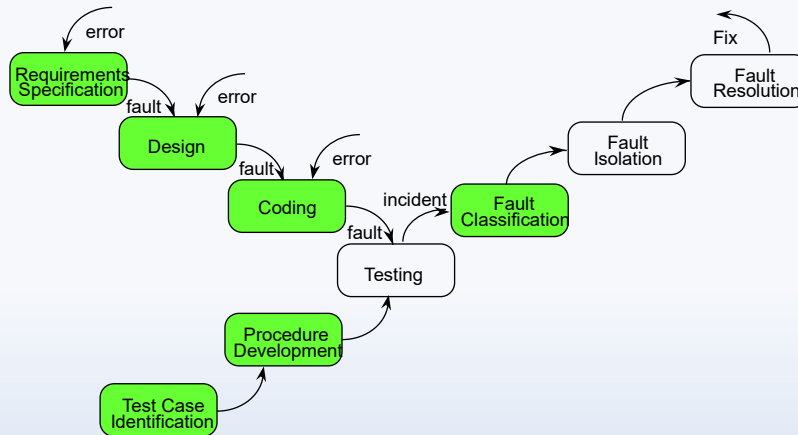
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Reasons to have Reviews

- **Communication** among designers
- **Training**, especially for new personnel
- Management progress **reporting**
- Defect **discovery**
- Performance **evaluation**
- Team **morale**

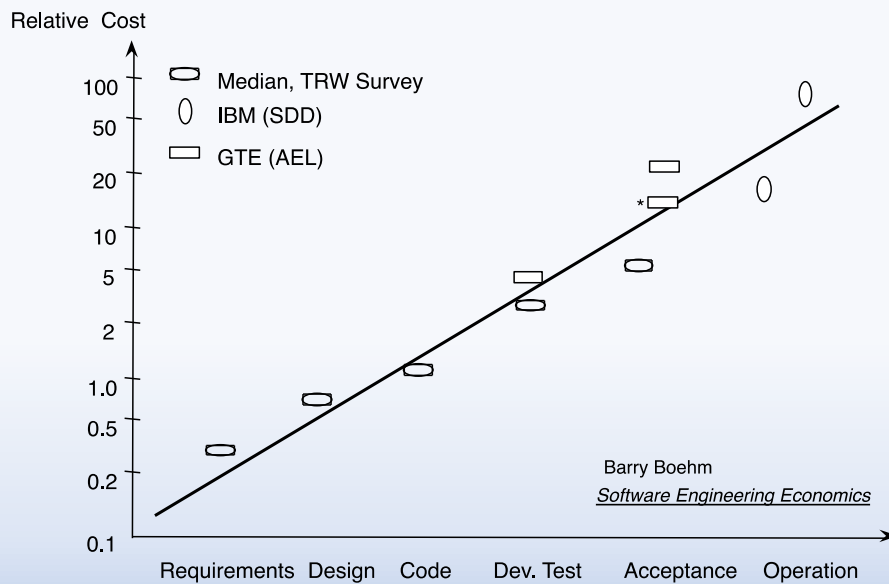
8

Reviews in the Software Lifecycle



9

Fault Resolution Costs



10

Cost/Benefit Reports

- Inspections can be 5% to 15% of total project cost
- “Jet Propulsion Laboratory estimated a net savings of \$7.5 million from 300 inspections performed on software they produced for NASA.”
- “another company reports annual savings of \$2.5 million”
 - Cost to fix a defect found by inspection: \$146
 - Cost to fix a defect found by customer: \$2900
 - Cost/Benefit ratio is 0.0503

Karl E. Wieggers “Improving Quality Through Software Inspections”

11

Review Roles & Purpose

- Roles
 - Responsible designer (aka Producer)
 - Recorder
 - Review leader
 - Reviewer
- To assure success, reviews must have
 - Reviewer credibility
 - Process credibility
 - Management and technical commitment

12

Review Roles

13

Role: Responsible Designer

- Individual (or group) whose work is being examined
- Role depends on type of review
- Responsible for resolving all action items.

14

Role: Recorder

- Not a secretary!
- Records information for Review Report
- Able to follow conversations and take notes
- Writes clearly and succinctly

15

Role: Review Leader

- Responsible for review
- Collects review materials
- Schedules review
- Conducts orderly, well-paced meeting*
- Writes review report with recorder

16

*Characteristics of a poor business meeting

- Participants find it to be a waste of time
- Wrong people in the meeting
- No agenda
- Little preparation
- No issues raised/identified
- Easily side-tracked
- Too much time spent solving problems
- Interruptions are acceptable

17

In Class Assignment: Fix These!

Issue	Resolution
Participants find it to be a waste of time	
Wrong people in the meeting	
No agenda	
Little preparation	
No issues raised/identified	
Easily side-tracked	
Too much time spent solving problems	
Interruptions are acceptable	

18

Attributes of a Good Review Leader

- Technically competent
- Strong Personality
 - A “natural” leader
 - Well-organized
 - A facilitator or mediator (referee?)
- Able to prioritize

19

Role: Reviewer

- Responsible for objectively reviewing the product
- Must be accountable for review
 - organization must define accountability
 - must have confidence in the review process
- Personality attributes
 - Objective
 - Unbiased
 - Technically competent

20

Multiple Review Roles?



- Should individuals be allowed to have multiple review roles?
- A. Yes, of course, from any of the roles.
- B. No, of course not!
- C. Sometimes it can be ok, in some specific cases.

Review Roles

- Responsible designer (aka Producer)
- Recorder
- Review leader
- Reviewer

21

Human Factors

- **Duration?** Maximum is 2 hours, 1 is better
- **Environment?** Quiet conference room with no interruptions
- **Best time?** 1 hour after work day begins
- **Worst time?** Friday after lunch (or maybe Saturday morning!)

22

Management Participation



Should management be involved in reviews?

- A. Yes, they should participate and review.
- B. Yes, they should participate, but not review.
- C. No, they shouldn't participate but should be informed of outcome.
- D. No, they shouldn't be involved in any way.

23

Management Participation in reviews? Bad Policy!

- Creates additional stress
- Conflict of interest
- Cannot be a spectator
- Evaluation of a product easily degenerates into evaluation of the producer
- Credibility as a reviewer?

24

Management Participation

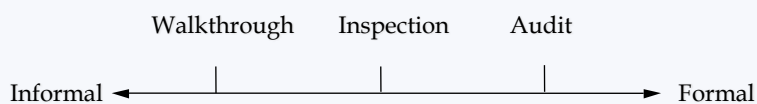
Would your management let you keep them out of a review?

- Yup, they trust me.
- Nope, they would insist on being there.

Possible compromise: earn right to participate by doing normal review preparation

25

Types of Reviews



- **Formality**

- Informal reviews often involve just 2 engineers, short duration, no documentation, low credibility
- formal reviews involve commitment, preparation and responsibility

- **Internal reviews:** very effective

- **External reviews:** carefully planned, rehearsed

- seldom very effective (at finding faults!)
- easily degenerate into Dog and Pony shows

26

Walkthroughs

- Responsible designer is review leader
- Most common form of review
- Most questions stimulated by presenter
- Frequently a presentation
- Effectiveness depends on...
 - real goal of responsible designer
 - preparation of reviewer(s)

27

Technical Inspections

- Responsible designer is not the review leader
- Formal process
- Very participative
- Advance preparation required
- Centers on checklist
- Effectiveness depends on ...
 - reviewer preparation
 - nature of checklist

28

Sidebar: Checklists

- Extremely valuable
- Organization dependent
- Refined by experience
- Not a secret! (at least not internally)
 - Helpful to designers
 - Essential to reviewers
- Often company confidential
 - (who wants to advertise their weaknesses?)

29

Walkthroughs vs Inspections

	Walkthrough	Inspection
Coverage	Broad, Superficial	Deep, Thorough
"Driver"	Responsible Designer	Review Leader
Preparation Time	Low	High
Formality	Informal	Formal
Effectiveness	Low	High

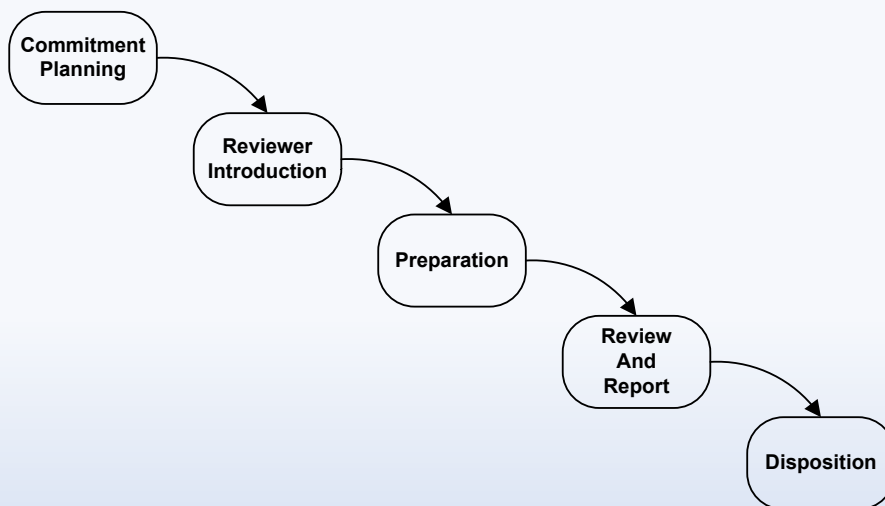
30

An Industrial-Strength Inspection Process

- Steps:
 1. Commitment planning
 2. Reviewer introduction
 3. Preparation
 4. Review meeting and report
 5. Disposition
- Other Factors
 - Review Etiquette
 - Accountability
 - Importance of public information

31

Industrial-Strength Inspection Phases



32

Commitment Planning

- Producer and supervisor **identify reviewers and review leader**.
- Identify the following:
 - Scope of review.
 - Commitments for participation, preparation, and meeting time.
 - Task charge accounting information.
- Communicate results to Review Leader.

33

Reviewer Introduction

- Producer **freezes deliverable item**, prepares reviewer packet with all pertinent information.
- Review Leader conducts preliminary meeting:
 - producer distributes reviewer packet.
 - producer gives brief overview of packet contents.
 - general discussion of major considerations ("soft" spots,).
 - revise checklist if necessary.
 - review roles determined and agreed upon.
- Obtain commitment from each reviewer for meeting.

34

Review Packet Contents

- Work product to be inspected
- Applicable standards (internal? External?)
- Checklist
- Definition of fault severity
- Blank issues list
- Individual ballot
 - Approve
 - Minor rework
 - Rework and review again

35

Preparation

- **Reviewers examine** review packet with respect to checklists and other concerns
- For each fault identified, **reviewers document** fault including severity
- Each reviewer **sends an "issues list"** to Review Leader ***well before*** the review meeting.
- Review Leader **prepares review meeting agenda** to reflect priorities of preliminary issues lists.
- Review Leader **distributes agenda** to review team prior to review meeting.

36

Review & Report

- First order of business: Postpone Review?
- Conduct review, following agenda of prioritized fault issues.
- Review Leader determines team consensus:
 - OK as is
 - OK with modifications (no re-review needed)
 - Reject: rework and re-review

37

Disposition

- Recorder prepares review report:
 - prioritized issues list
 - summary of individual ballots
- Review Leader distributes review report to:
 - review team
 - producer's supervisor
- Open items (i.e., action items) are resolved, and the review is closed.

38

Effective (& Ineffective) Review Culture

39

Review Etiquette

- Reviewers should be prepared.
- Review the product, not the producer.
- Stick to technical issues.
- Be constructive; reviews are not the place for
 - negative comments
 - "pats on the back"
- Identify issues, don't resolve them.
- Provide minor comments (e.g., spelling corrections) in writing to the producer at the end of the review meeting.
- Avoid discussions of style.
- Record all issues in public.
- Above all: Don't waste time!

40

Worst Case Review

- Producer picks friendly reviewers
- Short lead time
- No approved preparation time
- Deliverable not frozen
- Postponed twice
- Some reviewers absent
- Top designers excused (from reviews and reviewing)
- No checklist
- No issue list, no action items
- Page-by-page agenda
- Faults are resolved
- Coffee and lunch breaks needed
- Reviewers leave and return
- Producer's supervisor is review leader
- "cast of thousands"

41

Desirable Review Culture

- Producers do not dread reviews
- Reviewers have approved preparation time
- Materials delivered with sufficient lead time
- Trained review leaders
- Reviewers perceive reviews as productive
- Management perceives reviews as productive
- Review meetings have high priority
- Checklists are actively maintained
- Top designers are frequent reviewers
- Reviewer effectiveness is considered during performance evaluation
- Management actively supports reviews
- Management uses results of reviews

42

Discussion

- Should reviews in high-assurance fields match reviews in other fields?
- How could we maintain the benefits of a review process like this, without losing the effectiveness?